

* An equivalence relation $\sim$ $\{ \phi_i \}$

i. reflex.

ii. Symm

iii. transitive

Eq class: is a set:

$$[\phi] \equiv \{ \psi \mid \psi \sim \phi \}$$

$\Rightarrow$ In QM: $[\phi]$ eqv up to a set of measure zero in $L^2(\mathbb{R})$

$$\langle \psi, \psi \rangle = \int dx \psi^*(x) \psi(x) \rho(x)$$

$$\Rightarrow \psi \sim \phi \Rightarrow \psi = \phi + h$$

$$\langle h, h \rangle = 0 \Rightarrow \text{measure of } h \text{ is zero}$$

$$[\phi]:$$

$$\left[\begin{array}{l} L^2(\mathbb{R}) \text{ on } \psi \\ \hookrightarrow \overbrace{L^2(\mathbb{R})} \text{ on } [\phi] \end{array} \right] \hookrightarrow$$

Bundles or,
 $\mathbb{C}^1$ Bundles

Gauge: Sections of Princ. bundles

$$A \in \mathfrak{g} \otimes \underline{\Omega^1(M)}$$

$\Rightarrow \mathfrak{g}$ valued section

$$A = A^\alpha T^\alpha dx^\mu$$

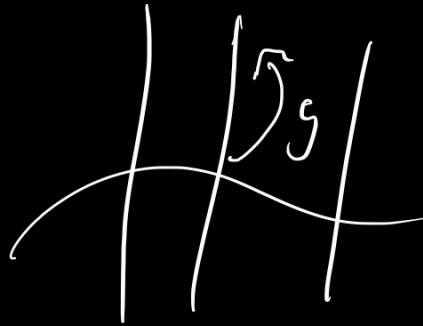
$$A = A_\mu T^\mu dx$$

$$\wedge B = B_\mu^\alpha T^\alpha dx^\mu$$

$$\Rightarrow A - B = dC$$

As 1 forms, form an eq. class
of Exact forms

→ In physical language: we can
move along the section by left action
of g on A
to get B



So the mapping onto the line is irrelevant

$$\begin{cases} \rightarrow [A] \\ \hookrightarrow [\psi] \end{cases}$$